

Distributed Integrated Control of a Mixed Traffic Network With Urban and Freeway Networks

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Abstract—In the optimization of mixed urban and freeway networks, most existing control strategies implement centralized controllers that are not appropriate for real-time control in large-scale traffic networks due to the huge computational complexity. In this article, a distributed control framework for the cooperative control of mixed urban and freeway traffic networks is developed. First, an integrated traffic model is presented to characterize the interactions between the urban and freeway networks. In addition, a partitioning method is proposed, and then, the mixed traffic network is divided among urban agents and freeway agents for the easy implementation of the distributed control framework. The optimization problem of the whole network can then be decomposed into a number of suboptimization problems based on the partition of the network. In the proposed distributed control strategy, each agent solves its own optimization problem independently with local information and transmitted information from neighboring agents to seek the Nash equilibrium. Finally, the computational efficiency and the effectiveness of the distributed integrated control strategy of mixed traffic networks are evaluated through a numerical example.

Index Terms—Distributed control, integrated control, mixed traffic networks, Nash equilibrium, urban and freeway agents.

I. INTRODUCTION

THE severe congestion due to the increasing traffic demand has led to significant economic loss and air pollution for metropolitan areas. Since freeways and urban networks in the metropolitan transportation network are interrelated and inherently coupled, the different traffic flow characteristics of the two networks call for an effective control approach for the mixed traffic network.

Currently, most optimization methods for traffic networks apply independent control for the urban and freeway parts of a transport network, ignoring the interrelation between the two entities. A few efforts in the literature have been contributed to the control of freeway networks. Ramp metering [1]–[4] employs traffic lights at the on-ramps to limit the entering flow into the mainstream freeway networks; variable speed

limit systems [5]–[8] are used to display variable speed signs in the road-side infrastructures to provide the advisory driving speed in response to the current traffic conditions; and route guidance systems [9]–[11] compute the alternative routes for drivers and are mostly useful in cases of incidents. Meanwhile, many researchers have aimed at the improvement of traffic congestion in urban networks. Traffic signal control [12]–[15] is the most commonly used method for the congestion alleviation. In addition, perimeter and boundary flow control strategies [16]–[18] regulate the input flow to the traffic network, so as to hold vehicles outside the controlled area.

However, because of the inherent couplings between urban and freeway networks, traffic management systems based on the independent considerations of each subsystem will lead to a competitive result. In the case of heavy congestion in urban networks, the continuous traffic demand from freeways and the limited space on off-ramps and connected downstream intersection in urban networks will escalate the existing congestion, leading to spillover to the mainstream of freeway networks. In addition, the capacity (the maximum outflow) of the freeway network will drop accordingly due to the lane reduction caused by the spillover, and the breakdown areas will then extend to further freeway links. For freeway networks, any control strategy to reduce travel time in the network may impede the entering flow from urban networks and maintain high outflows to the city, which may create long queues at both on-ramps and off-ramps.

A few efforts have been devoted to the integrated control of mixed traffic networks [19]–[22]. An integrated traffic-responsive controller has been studied by [20], where a local feedback ramp metering strategy is derived for the freeway roads and a linear-quadratic optimal controller is utilized in the urban network. In addition, an integrated model predictive controller that optimizes traffic light signals for urban networks and variable speed limit signs for freeway networks has been investigated in [21]. Moreover, perimeter controllers have been operated to manipulate the transfer flow between the urban regions, and ramp metering controllers have been considered to coordinate the flow distribution in the mixed network in [22].

Nevertheless, the aforementioned control approaches for mixed traffic networks all apply centralized control (CC). The computational burden introduced by the complexity of traffic dynamics in the mixed network makes it infeasible for real-world applications. On the other hand, the hazardous spill-back effects at off-ramps have not been paid enough attention by most studies.

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In this article, the distributed model predictive control approach of traffic signals proposed in [23] is extended to involve the control of the freeway part in the mixed traffic network. The variable speed limits of freeway links and the traffic signals of intersections are optimized in a coordinated way to manipulate the traffic flow in the whole network. The integrated control of a mixed urban and freeway network calls for a traffic flow model that can describe the traffic dynamics of the urban and freeway networks, as well as the interrelations between them. In addition, a centralized controller is not appropriate for such a large-scale traffic network because of the huge communicational and computational burden. Therefore, in the modeling part, the couplings between the urban and freeway parts are modeled by the traffic dynamics of on-ramps and off-ramps. Specifically, the spill-back effects of off-ramps are developed. In the design of the control scheme, a distributed control framework is proposed. The mixed traffic network is divided among agents, in which the control actions of each agent are determined in a parallel way based on their local information and the transmitted information from their neighbors. Due to the distributed architecture, the proposed control approach has a very low computational demand in each subsystem, making it easily implemented in large-scale traffic networks.

The outline of the remainder is as follows. In Section II, the traffic dynamics of urban and freeway networks are presented, and the spill-back effects of off-ramps are modeled in the extended METANET model. In Section III, the optimization problem of a mixed traffic network is formulated based on the model predictive control approach. Afterward, the partitioning method of the network and the corresponding multiagent control problem formulation are developed in Section IV. Furthermore, the computation and communication procedures of each agent are also described. The effectiveness of the proposed control scheme is explored in Section V by case studies. Section VI concludes this article.

II. TRAFFIC MODELING FOR MIXED TRAFFIC NETWORKS

In this article, the S model [24] and the nondestination-oriented METANET model [25] are selected as the prediction model to characterize the traffic dynamics of the urban and freeway networks, respectively. They are also used in earlier work [21], [26]–[30] for the independent control of the urban and freeway networks. Note that other macroscopic traffic models that depict the dynamics of the traffic network can also be used.

A. Urban Traffic Modeling

An urban network consists of intersections and roads, which can be denoted by the set of nodes $\mathcal{V}$ and the set of links $\mathcal{E}$ in a directed graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$. The road departing from intersection $u \in \mathcal{V}$ toward intersection $d \in \mathcal{V}$ is denoted by an ordered pair $(u, d) \in \mathcal{E}$.

In the urban network, the green time duration of traffic signals at intersections is used as control signals in this article. We assume that the cycle time, the offsets, and the phase sequence in each intersection have been obtained by

TABLE I
NOTATIONS IN THE URBAN TRAFFIC MODEL

T_u	sampling time interval in the urban network
k_u	time step counter of the urban network
$n_{u,d}(k)$	number of vehicles on link (u, d) at time step k
$l_{u,d}(k)$	queue length of on link (u, d) at time step k ($l_{u,d,o}(k)$ is the queue length of link (u, d) towards o)
$I_{u,d}(O_{u,d})$	set of input (output) nodes of link (u, d)
$\alpha_{u,d}^{\text{enter}}(k)$	average entering flow rate of link (u, d) at time step k
$\alpha_{u,d}^{\text{arrive}}(k)$	average arriving flow rate at the end of queue of link (u, d) at time step k ($\alpha_{u,d,o}^{\text{arrive}}(k)$ is the arriving flow rate of link (u, d) towards o)
$\alpha_{u,d}^{\text{leave}}(k)$	average leaving flow rate of link (u, d) at time step k ($\alpha_{u,d,o}^{\text{leave}}(k)$ is the leaving flow rate of (u, d) towards o)
$\beta_{u,d,o}(k)$	proportion of the traffic turning from (u, d) towards o at time step k
$g_{u,d,o}(k)$	green time duration in which traffic stream (u, d, o) have right of way during time step k
$G_{\max}$	maximum green time of each phase of the traffic signals
$G_{\min}$	minimum green time of each phase of the traffic signals
$Q_{u,d}$	saturated flow rate of traffic on link (u, d)
$v_{u,d}^{\text{free}}$	free flow speed of vehicles on link (u, d)
$C_{u,d}$	storage capacity of link (u, d)
$\lambda_{u,d}$	number of lanes of link (u, d)
l_{veh}	average vehicle length
$ \mathcal{V} $	number of intersections in the urban network

other control schemes. Besides, the cycle time is T_u for all intersections. To describe the S model, we first define the variables in Table I.

In the S model [24], the traffic state of link (u, d) is described by the number of vehicles on it, which is updated by

$$n_{u,d}(k_u + 1) = n_{u,d}(k_u) + (\alpha_{u,d}^{\text{enter}}(k_u) - \alpha_{u,d}^{\text{leave}}(k_u)) \cdot T_u \quad (1)$$

where $\alpha_{u,d}^{\text{enter}}(k_u)$ and $\alpha_{u,d}^{\text{leave}}(k_u)$ of link (u, d) can be calculated by

$$\begin{aligned} \alpha_{u,d}^{\text{enter}}(k_u) &= \sum_{i \in I_{u,d}} \alpha_{i,u,d}^{\text{leave}}(k_u) \\ \alpha_{u,d}^{\text{leave}}(k_u) &= \sum_{o \in O_{u,d}} \alpha_{u,d,o}^{\text{leave}}(k_u). \end{aligned} \quad (2)$$

The average leaving flow rate $\alpha_{u,d,o}^{\text{leave}}(k_u)$ is computed by

$$\begin{aligned} \alpha_{u,d,o}^{\text{leave}}(k_u) &= \min \left(\beta_{u,d,o}(k_u) \cdot Q_{u,d} \cdot g_{u,d,o}(k_u) / T_u, \right. \\ &\quad \left. l_{u,d,o}(k_u) / T_u + \alpha_{u,d,o}^{\text{arrive}}(k_u), \right. \\ &\quad \left. \frac{\beta_{u,d,o}(k_u)}{\sum_{i \in I_{d,o}} \beta_{i,d,o}(k_u)} (C_{d,o} - n_{d,o}(k_u)) / T_u \right). \end{aligned} \quad (3)$$

The queue length $q_{u,d,o}(k_u)$ of the traffic stream (u, d, o) is updated by

$$l_{u,d,o}(k_u + 1) = l_{u,d,o}(k_u) + (\alpha_{u,d,o}^{\text{arrive}}(k_u) - \alpha_{u,d,o}^{\text{leave}}(k_u)) \cdot T_u \quad (4)$$

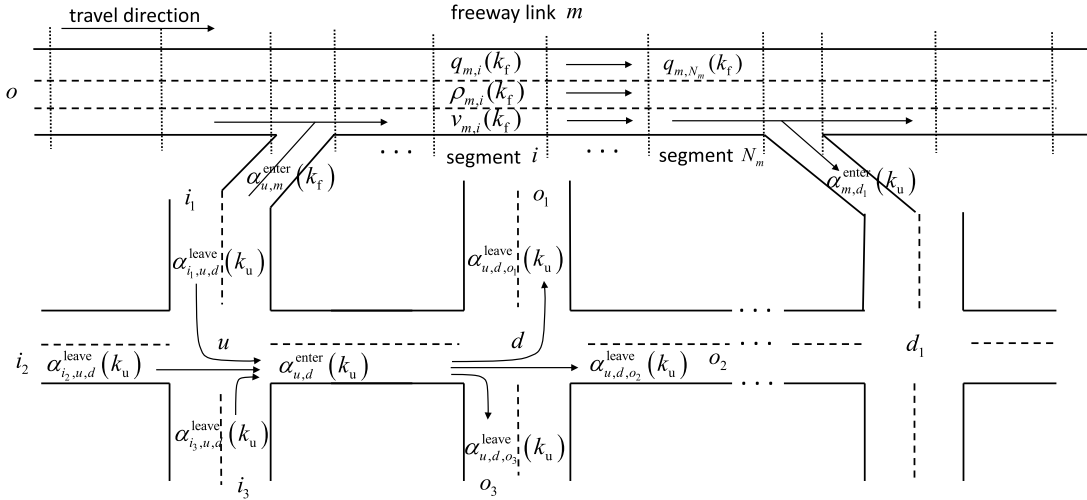


Fig. 1. Mixed traffic network consisting of urban and freeway networks.

where $\alpha_{u,d,o}^{\text{arrive}}(k_u)$ is given by

$$\begin{aligned} \alpha_{u,d}^{\text{arrive}}(k_u) &= (1 - \gamma_{u,d}(k_u)/T_u) \cdot \alpha_{u,d}^{\text{enter}}(k_u - \tau_{u,d}(k_u)) \\ &\quad + \gamma_{u,d}(k_u)/T_u \cdot \alpha_{u,d}^{\text{enter}}(k_u - \tau_{u,d}(k_u) - 1) \\ \phi_{u,d}(k_u) &= (C_{u,d} - l_{u,d}(k_u)) \cdot l_{\text{veh}} / (\lambda_{u,d} \cdot v_{u,d}^{\text{free}}) \\ \tau_{u,d}(k_u) &= \left\lfloor \frac{\phi_{u,d}(k_u)}{T_u} \right\rfloor, \quad \gamma_{u,d}(k_u) = \phi_{u,d}(k_u) - \tau_{u,d}(k_u) \cdot T_u \end{aligned} \quad (5)$$

with $\phi_{u,d}(k_u)$ being the travel time required for a vehicle to reach the end of the queue of link (u, d) , and $\lfloor x \rfloor$ denoting the maximum integer not larger than x .

Therefore, the S model that depicts the urban traffic characteristics can be represented by (1)–(5). By using the transformation and equivalences given in [26], it can be transformed to a mixed-integer model. The details of the transformation process can be referred to [31].

B. Freeway Traffic Modeling

A freeway network can be denoted by a set of freeway links $\mathcal{L}$. Moreover, the external traffic demand is generated from the origin link $o \in \mathcal{L}_{\text{org}}$. In the freeway network, the variable speed limits of links are used as control signals in this article.

In the METANET model, each freeway link $m \in \mathcal{L}$ is further divided into N_m segments with equal length L_m and has the same number of lanes λ_m , as shown in Fig. 1. The traffic density $\rho_{m,i}(k_f)$ and the mean speed $v_{m,i}(k_f)$ are used to depict the traffic dynamics of freeway segment (m, i) , and the number of vehicles $n_o(k_f)$ is used to depict the traffic dynamics of origin link o . The notations used in the freeway traffic model are listed in Table II.

1) *Motorway Links*: For freeway segment (m, i) , the basic equations to update $\rho_{m,i}(k_f)$ and $v_{m,i}(k_f)$ are given as follows:

$$\begin{aligned} \rho_{m,i}(k_f + 1) &= \rho_{m,i}(k_f) + \frac{T_f}{L_m \lambda_m} (q_{m,i-1}(k_f) - q_{m,i}(k_f)) \\ v_{m,i}(k_f + 1) &= v_{m,i}(k_f) + \frac{T_f}{\tau} (V(\rho_{m,i}(k_f)) - v_{m,i}(k_f)) \end{aligned}$$

$\mathcal{N}_i^u$	set of the neighboring agents in the urban network
$\mathcal{N}_i^f$	set of the neighboring agents in the freeway network
$n_i(k)$	vector consisting of the number of vehicles of roads in urban agent i at time step k
$l_i(k)$	vector consisting of the queue lengths of roads in urban agent i at time step k
$\alpha_i^{\text{enter}}(k)$	vector consisting of the entering flow rate of roads in urban agent i at time step k
$\alpha_i^{\text{arrive}}(k)$	vector consisting of the arriving flow rate of the traffic streams in urban agent i at time step k
$\alpha_i^{\text{leave}}(k)$	vector consisting of the leaving flow rate of the traffic streams in urban agent i at time step k
$d_i(k)$	external traffic demand of roads in urban agent i at time step k
$q_p(k)$	vector consisting of outflow of links in freeway agent p at time step k
$x_i^u(k)$	traffic state of urban agent i at time step k
$u_i^u(k)$	control signal of urban agent i at time step k
$y_{ji}^u(k)$	interaction variables between urban agent j and agent i at time step k
$y_{pi}^f(k)$	actual interaction variables between freeway agent p and agent i at time step k
$\rho_p(k)$	vector consisting of the traffic density of freeway segments in agent p at time step k
$v_p(k)$	vector consisting of the traffic speed of freeway segments in agent p at time step k
$v_p^{\text{lim}}(k)$	vector consisting of the speed limits of freeway links in agent p at time step k
$x_p^f(k)$	traffic state of freeway agent p at time step k
$u_p^f(k)$	control signal of freeway agent p at time step k
$\tilde{z}_{ji}(k)$	estimated interaction variables between agent j and i over the prediction horizon at time step k

$$\begin{aligned} &+ \frac{T_f}{L_m} v_{m,i}(k_f) (v_{m,i-1}(k_f) - v_{m,i}(k_f)) \\ &- \frac{v \cdot T_f}{\tau \cdot L_m} \frac{\rho_{m,i+1}(k_f) - \rho_{m,i}(k_f)}{\rho_{m,i}(k_f) + \kappa} \end{aligned} \quad (6)$$

where

$$q_{m,i}(k_f) = \lambda_m \cdot \rho_{m,i}(k_f) \cdot v_{m,i}(k_f). \quad (7)$$

The time constant τ , the anticipation constant v , and parameter κ are constants.

In addition, $V(\rho_{m,i}(k_f))$ describes the relationship between the desired speed and $\rho_{m,i}(k_f)$ of segment (m, i) according to the fundamental diagram. It can be given by

$$V(\rho_{m,i}(k_f)) = \min \left\{ v_m^{\text{free}} \exp \left[\frac{-1}{a_m} \left(\frac{\rho_{m,i}(k_f)}{\rho_m^{\text{crit}}(k_f)} \right)^{a_m} \right], (1 + \alpha) v_m^{\text{lim}}(k_f) \right\} \quad (8)$$

where a_m is a parameter for the fundamental diagram. It is noticed that the segments of link m have the same speed limit.

Specifically, some modifications are made to (6) when the freeway link has merging or diverging behaviors.

If freeway link m has more than one entering link (such as on-ramps), $v_{m,i-1}(k_f)$ in (6) with $i = 1$ is computed by

$$v_{m,0}(k_f) = \frac{\sum_{l \in I_m} v_{l,N_l}(k_f) \cdot q_{l,N_l}(k_f)}{\sum_{l \in I_m} q_{l,N_l}(k_f)} \quad (9)$$

where I_m is the set of entering links of link m .

If freeway link m has more than one leaving link (such as off-ramps), $\rho_{m,i+1}(k_f)$ in (6) with $i = N_m$ is computed by

$$\rho_{m,N_m+1}(k_f) = \frac{\sum_{l \in O_m} \rho_{l,1}^2(k_f)}{\sum_{l \in O_m} \rho_{l,1}(k_f)} \quad (10)$$

where O_m is the set of leaving links of link m .

2) *Origin Links*: For origin link $o \in \mathcal{L}_{\text{org}}$ of the freeway network, a queue model is used. The number of vehicles on link o is updated by

$$n_o(k_f + 1) = n_o(k_f) + (d_o(k_f) - q_o(k_f)) \cdot T_f. \quad (11)$$

The outflow $q_o(k_f)$ can be calculated by

$$q_o(k_f) = \min \left\{ d_o(k_f) + \frac{n_o(k_f)}{T_f}, q_o^{\text{max}}(k_f) \right\} \quad (12)$$

where $q_o^{\text{max}}(k_f)$ is given by

$$q_o^{\text{max}}(k_f) = \begin{cases} Q_o, & \text{if } \rho_{l,1}(k_f) < \rho_l^{\text{crit}} \\ Q_o \frac{\rho_l^{\text{max}} - \rho_{l,1}(k_f)}{\rho_l^{\text{max}} - \rho_l^{\text{crit}}}, & \text{else} \end{cases} \quad (13)$$

with l denoting the downstream freeway link of link o .

C. Interface Between the Urban and Freeway Networks

The interconnected roads between the urban and freeway parts of the mixed traffic network include on-ramps and off-ramps, which are denoted by $\mathcal{E}_{\text{on}} = \{(u, m) | u \in \mathcal{V} \text{ and } m \in \mathcal{L}\}$ and $\mathcal{E}_{\text{off}} = \{(m, d) | m \in \mathcal{L} \text{ and } d \in \mathcal{V}\}$, respectively. The traffic dynamics of on-ramps and off-ramps are both described by queue models.

1) *On-Ramps*: According to the conservation law of traffic flow, the number of vehicles on on-ramp $(u, m) \in \mathcal{E}_{\text{on}}$ can be updated by

$$n_{u,m}(k_f + 1) = n_{u,m}(k_f) + (\alpha_{u,m}^{\text{enter}}(k_f) - \alpha_{u,m}^{\text{leave}}(k_f)) \cdot T_f. \quad (14)$$

For on-ramp (u, m) , the entering vehicles are controlled by the upstream urban intersection. Considering the different sampling time intervals of the urban and freeway networks, the entering flow rate of on-ramp (u, m) is synchronized with the leaving flow rate of upstream urban roads.

We define the continuous leaving flow rate of (i, u, m) as

$$\alpha_{i,u,m}^{\text{leave, cont}}(t) = \alpha_{i,u,m}^{\text{leave}}(k_u), \quad k_u T_u \leq t < (k_u + 1) T_u$$

where $\alpha_{i,u,m}^{\text{leave}}(k_u)$ is given by (3). Then, the entering flow rate $\alpha_{u,m}^{\text{enter}}(k_f)$ can be computed by

$$\alpha_{u,m}^{\text{enter}}(k_f) = \frac{1}{T_f} \sum_{i \in I_{u,m}} \int_{k_f T_f}^{(k_f+1) T_f} \alpha_{i,u,m}^{\text{leave, cont}}(t) dt. \quad (15)$$

Moreover, $\alpha_{u,m}^{\text{leave}}(k_f)$ is computed in a similar way with (12), which can be given by

$$\alpha_{u,m}^{\text{leave}}(k_f) = \min \left\{ \alpha_{u,m}^{\text{enter}}(k_f) + \frac{n_{u,m}(k_f)}{T_f}, q_{u,m}^{\text{max}}(k_f) \right\} \quad (16)$$

where $q_{u,m}^{\text{max}}(k_f)$ can be computed by (13) with o replaced by (u, m) .

Therefore, the traffic dynamics of on-ramps are influenced by the traffic density of the downstream freeway link. It will further put an impact on the traffic states of upstream urban roads of on-ramps, reflecting the interactions between the urban and freeway networks.

2) *Off-Ramps*: The number of vehicles on off-ramp (m, d) is updated by

$$n_{m,d}(k_u + 1) = n_{m,d}(k_u) + (\alpha_{m,d}^{\text{enter}}(k_u) - \alpha_{m,d}^{\text{leave}}(k_u)) \cdot T_u. \quad (17)$$

For off-ramp (m, d) , the leaving flow rate $\alpha_{m,d}^{\text{leave}}(k_u)$ is controlled by the traffic lights of the downstream intersection and can be computed by (3). Moreover, the entering flow rate $\alpha_{m,d}^{\text{enter}}(k_u)$ is determined by the outflow of the upstream freeway segment (m, N_m)

$$\alpha_{m,d}^{\text{enter}}(k_u) = \beta_{m,d}(k_u) q_{m,N_m}(k_u) \quad (18)$$

where $\beta_{m,d}(k_u)$ is the proportion of the traffic leaving freeway link m toward intersection d .

In (18), $q_{m,N_m}(k_u)$ denotes the average outflow of (m, N_m) during urban time step k_u . We define the continuous outflow of the freeway segment (m, N_m) as

$$q_{m,N_m}^{\text{cont}}(t) = q_{m,N_m}(k_f), \quad k_f T_f \leq t < (k_f + 1) T_f$$

where $q_{m,N_m}(k_f)$ is given as

$$q_{m,N_m}(k_f) = \lambda_{m,N_m}(k_f) \rho_{m,N_m}(k_f) v_{m,N_m}(k_f). \quad (19)$$

Then, the outflow $\alpha_{m,d}^{\text{enter}}(k_u)$ at urban time step k_u can be computed by

$$q_{m,N_m}(k_u) = \frac{1}{T_u} \int_{k_u T_u}^{(k_u+1) T_u} q_{m,N_m}^{\text{cont}}(t) dt. \quad (20)$$

Since link m has more than one leaving link: off-ramp (m, d) and the downstream freeway link $m + 1$, $v_{m,N_m}(k_f)$ is influenced by the traffic density $\rho_{m,d}(k_f)$ and the traffic density $\rho_{m+1,1}(k_f)$, as implied in (6) and (10), where $\rho_{m,d}(k_f)$ can be computed by

$$\rho_{m,d}(k_f) = \frac{n_{m,d}(k_f)}{\lambda_{m,d} L_{m,d}}. \quad (21)$$

D. Effects of Off-Ramps

According to (6), (10), and (17)–(21), the traffic dynamics of off-ramp (m, d) is influenced by the outflow of its upstream freeway segment (m, N_m) . On the other hand, the traffic dynamics of (m, N_m) is in turn influenced by the number of vehicles on link (m, d) . However, the spill-back effects of off-ramps are not directly reflected in the existing traffic model.

When the congestion of (m, d) spreads to the upstream freeway link, the vehicles on (m, N_m) will be impeded, resulting in a standing queue on (m, N_m) and a reduction of the lane number of (m, N_m) . Moreover, it will further reduce the outflow q_{m,N_m} and increase the traffic density ρ_{m,N_m} . In the METANET model, $q_{m,N_m}(k_f)$ is only determined by $\lambda_{m,N_m}(k_f)$, $\rho_{m,N_m}(k_f)$, and $v_{m,N_m}(k_f)$, as implied by (19), without the consideration of traffic condition of off-ramp (m, d) . Thus, we extend the METANET model in this article to incorporate the spill-back effects of off-ramps.

Here, we define the lane number $\lambda_{m,N_m}(k_f)$ of (m, N_m) as a function of the number of vehicles of (m, d) on the freeway segment (m, N_m)

$$\lambda_{m,N_m}(k_f) = (\lambda_m - 1) + \exp \left[-\frac{1}{b} \left(\frac{n_{m,d}(k_f)}{\gamma_{m,d} C_{m,d}} \right)^b \right] \quad (22)$$

where λ_m represents the original lane number of (m, N_m) , $\gamma_{m,d}$ and b are suitably defined parameters such that whenever the number of vehicles $n_{m,d}(k_f)$ exceeds a certain threshold, and $\lambda_{m,N_m}(k_f)$ will be reduced from λ_m to $\lambda_m - 1$ to characterize the spill-back effects of off-ramp (m, d) . After the elimination of the congestion on (m, d) , $\lambda_{m,N_m}(k_f)$ and the corresponding flow capacity of (m, N_m) will be recovered. Note, however, that other methods that describe the interrelations between the number of vehicles of off-ramps and the traffic conditions of its upstream freeway links can also be used.

Therefore, through (22), the interactions between the urban and freeway networks can be directly depicted. In addition, when the congestion on (m, N_m) is continuously propagated to the upstream freeway links, the lane number and the outflow capacity of the affected links will also be reduced. The congestion propagation in the freeway network and the interrelations between the freeway links is characterized by the state equations of the METANET model.

III. INTEGRATED CONTROL OF MIXED TRAFFIC NETWORKS

The couplings between the urban and freeway parts in the mixed traffic network prompt the necessity of the integrated control of the two parts. In this article, a model predictive control strategy is applied for the coordinated control of the

mixed traffic network to deal with the uncertainty and the multiobjective requirements in the optimization process. By regulating the traffic signals of intersections and the variable speed limits of freeway links, the cost function over the prediction horizon is minimized. The traffic model described in Section II is used as the prediction model. Note, however, that the proposed control approach also works with other traffic flow models.

We define the least common multiple of the urban sampling time interval T_u and the freeway sampling time interval T_f as

$$T_{uf} = [T_u, T_f]$$

with $[a, b]$ denoting the least common multiple of a and b . Then, the control interval T_c is defined as the integer multiple of T_{uf}

$$T_c = N_c T_{uf} \quad (23)$$

where N_c is an integer. Therefore, the control signals in the mixed traffic network are updated every T_c s.

A. Objective Function

We choose the total time spent (TTS) of vehicles in the whole network and the penalization of large variations of control signals as the objective function of the integrated control of a mixed traffic network. In the TTS term, the number of vehicles is counted for origin links, mainstream freeway links, urban roads, on-ramps, and off-ramps. In the penalization term, the variations of the variable speed limits and the green time duration of traffic lights are considered. Note, however, that the proposed approach also works equally well for other objective functions that describe the network performance.

The optimization problem of the mixed traffic network is solved at each control time step k_c , with the objective function computed over the prediction horizon $t \in [k_c T_c, (k_c + N_p) T_c]$. Let $k_c T_c = k_u^0 T_u = k_f^0 T_f$ and $(k_c + N_p) T_c = (k_u^1 + 1) T_u = (k_f^1 + 1) T_f$; then, the objective function can be written as

$$\begin{aligned} J(k_c) = & T_u \sum_{k=k_u^0}^{k_u^1} \sum_{(u,d) \in (\mathcal{E} \cup \mathcal{E}_{\text{off}})} n_{u,d}(k+1) \\ & + T_f \sum_{k=k_f^0}^{k_f^1} \left\{ n_o(k+1) + \sum_{(u,m) \in \mathcal{E}_{\text{on}}} n_{u,m}(k+1) \right. \\ & \quad \left. + \sum_{m \in \mathcal{L}} \sum_{i=1}^{N_m} \rho_{m,i}(k+1) L_m \lambda_m \right\} \\ & + \sum_{k=k_c}^{k_c+N_p-2} (u(k+1) - u(k))^T R (u(k+1) - u(k)) \end{aligned} \quad (24)$$

where $u = [v_1^{\text{lim}} \cdots v_{|\mathcal{L}|}^{\text{lim}} g_1^T \cdots g_{|\mathcal{V}|}^T]^T$ represents the control signals of the mixed traffic network and g_i denotes the vector consisting of the green time duration of phases of traffic signals in intersection i . Moreover, R is a weight matrix reflecting the tradeoff between the TTS term and the control signal variations' penalty term.

B. Constraints

Because of physical limitations in the traffic network, the number of vehicles and queue lengths of the urban roads, on-ramps, and off-ramps, and the traffic density and the traffic speed of freeway links are restricted inside the corresponding feasible sets

$$0 \leq n_{u,d}(k_u) \leq C_{u,d}, \quad 0 \leq l_{u,d}(k_u) \leq C_{u,d} \quad (25)$$

$$0 \leq \rho_{m,i}(k_f) \leq \rho_m^{\max}, \quad 0 \leq v_{m,i}(k_f) \leq v_m^{\text{free}} \quad (26)$$

where $(u, d) \in \mathcal{E} \cup \mathcal{E}_{\text{on}} \cup \mathcal{E}_{\text{off}}$, $m \in \mathcal{L}$, $i = 1, \dots, N_m$.

Moreover, due to the restrictions on actuators, the green time duration of each phase of the traffic signals and the variable speed limits of freeway links are supposed to satisfy certain constraints. They are all restricted by lower and upper bounds. Besides, for safety considerations, the maximum decrease in the speed limit of one link during consecutive time steps and the maximum decrease in the speed limits of two consecutive links at the same time step are limited by $\Delta v_{\max}$. The above constraints can be given as

$$G_{\min} \leq g_i(k_c) \leq G_{\max}, \quad \text{for } i \in \mathcal{V} \quad (27)$$

$$\begin{aligned} v_{\min}^{\text{lim}} &\leq v_m^{\text{lim}}(k_c) \leq v_m^{\text{free}} \\ v_m^{\text{lim}}(k_c) - v_m^{\text{lim}}(k_c - 1) &\geq -\Delta v_{\max} \\ v_m^{\text{lim}}(k_c) - v_{m-1}^{\text{lim}}(k_c) &\geq -\Delta v_{\max}, \quad \text{for } m \in \mathcal{L}. \end{aligned} \quad (28)$$

To reduce the delay time of vehicles caused by speed limits, there are no restrictions on the variations of VSL when it is increased between two consecutive control time steps or when the speed limit of the freeway link is increased with respect to its upstream link. The restrictions on the variation of VSL are on one side.

C. Optimization Problem Formulation

Above all, the optimization problem of the integrated control of mixed traffic networks MPC can be formulated as

$$\begin{aligned} \min_{\tilde{u}(k_c)} \quad & J(k_c) \\ \text{s.t.} \quad & (1) - (22), \text{ and } (25) - (28) \end{aligned}$$

where $\tilde{u}(k_c) = [u(k_c), u(k_c + 1), \dots, u(k_c + N_p - 1)]^T$ represents the control sequence obtained by solving the optimization problem at time step k_c .

The optimization problem involves a great number of control variables and constraints, which results in large communicational and computational complexities, making it not appropriate for online optimization. Therefore, a distributed control scheme is proposed in this article.

IV. DISTRIBUTED CONTROLLER DESIGN

To apply the distributed control scheme, the mixed traffic network is first partitioned among agents. Then, the overall optimization problem is divided into a number of suboptimization problems, which is solved by each agent in parallel. The notations used in the section are listed in Table III.

TABLE III
NOTATIONS UNDER THE DISTRIBUTED CONTROL ARCHITECTURE

$\mathcal{N}_i^u$	set of the neighboring agents in the urban network
$\mathcal{N}_i^f$	set of the neighboring agents in the freeway network
$n_i(k)$	vector consisting of the number of vehicles of roads in urban agent i at time step k
$l_i(k)$	vector consisting of the queue lengths of roads in urban agent i at time step k
$\alpha_i^{\text{enter}}(k)$	vector consisting of the entering flow rate of roads in urban agent i at time step k
$\alpha_i^{\text{arrive}}(k)$	vector consisting of the arriving flow rate of the traffic streams in urban agent i at time step k
$\alpha_i^{\text{leave}}(k)$	vector consisting of the leaving flow rate of the traffic streams in urban agent i at time step k
$d_i(k)$	external traffic demand of roads in urban agent i at time step k
$q_p(k)$	vector consisting of outflow of links in freeway agent p at time step k
$x_i^u(k)$	traffic state of urban agent i at time step k
$u_i^u(k)$	control signal of urban agent i at time step k
$y_{ji}^u(k)$	interaction variables between urban agent j and agent i at time step k
$y_{pi}^f(k)$	actual interaction variables between freeway agent p and agent i at time step k
$\rho_p(k)$	vector consisting of the traffic density of freeway segments in agent p at time step k
$v_p(k)$	vector consisting of the traffic speed of freeway segments in agent p at time step k
$v_p^{\text{lim}}(k)$	vector consisting of the speed limits of freeway links in agent p at time step k
$x_p^f(k)$	traffic state of freeway agent p at time step k
$u_p^f(k)$	control signal of freeway agent p at time step k
$\tilde{z}_{ji}(k)$	estimated interaction variables between agent j and i over the prediction horizon at time step k

A. Network Partition

1) *Urban Agent*: The partitioning method of the urban network is the same as that in [23]. The set of urban agents is denoted by $\mathcal{A}_u$, where agent i is comprised of intersection i and its incoming links. Note that off-ramp (m, i) is also contained in agent i .

After the partition of the traffic network, the neighboring agents can then be defined. The neighbors of agent i consist of urban agents and freeway agents

$$\mathcal{N}_i = \mathcal{N}_i^u \cup \mathcal{N}_i^f.$$

Urban agent j is a neighbor of urban agent i if there is a road from intersection j to i . Freeway agent p is a neighbor of agent i if there is an on-ramp originating from intersection i toward one of the freeway links in p or there is an off-ramp originating from one of the freeway links in p toward intersection i . For instance, in Fig. 2, the set of neighboring urban agents of i_1 is $\mathcal{N}_{i_1}^u = \{i_2, i_3\}$, and the set of neighboring freeway agents of i_1 is $\mathcal{N}_{i_1}^f = \{p - 1\}$.

2) *Freeway Agent*: The set of freeway agents is denoted by $\mathcal{A}_f$, where agent p is comprised of a set of consecutive freeway links and the adjacent on-ramps such that at most one off-ramp is connected to the links in p . Moreover, the origin link of the freeway network is also contained in p if it is adjacent to the links in p . The set of freeway links in agent p is denoted by $\mathcal{L}_p$. The partitioning results of a freeway network are illustrated in Fig. 3.

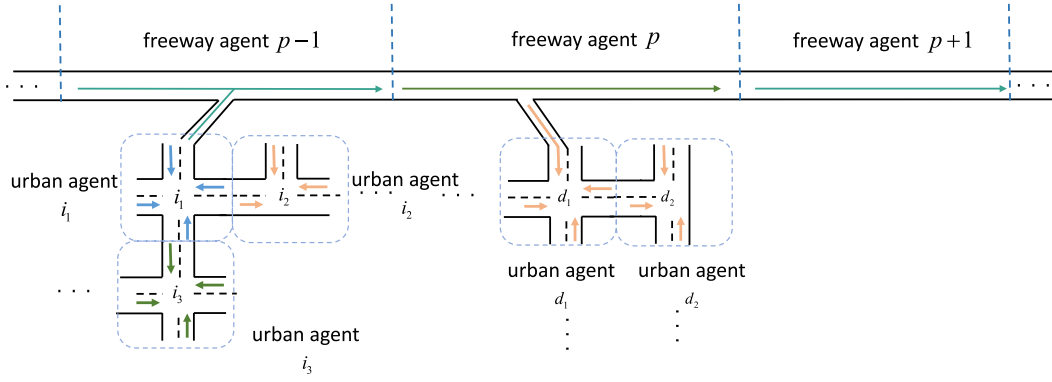


Fig. 2. Urban and freeway agents in a mixed traffic network.

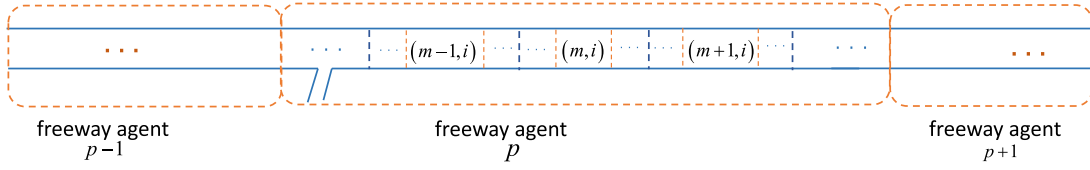


Fig. 3. Freeway agents.

Freeway agent h is a neighbor of freeway agent p if one of the links in h is the upstream or the downstream link of one of the links in p . Urban agent i is a neighbor of p if p is one of the neighbors of i . In Fig. 2, the set of neighboring freeway agents of agent p is $\mathcal{N}_p^f = \{p-1, p+1\}$, and the set of neighboring urban agents of agent p is $\mathcal{N}_p^u = \{d_1\}$.

B. Agent-Based Network Model

Based on the partitioning method of the mixed traffic network, the traffic model of each agent can then be obtained.

1) *Traffic Flow Model of an Urban Agent*: The traffic model of an urban agent is extended from [31]. The number of vehicles and queue lengths of urban agent $i \in \mathcal{A}_u$ at time step $k_u + 1$ can be given as

$$\begin{aligned} n_i(k_u + 1) &= n_i(k_u) + T_u \cdot (\alpha_i^{\text{enter}}(k_u) - \alpha_i^{\text{leave}}(k_u)) \\ l_i(k_u + 1) &= l_i(k_u) + T_u \cdot (\alpha_i^{\text{arrive}}(k_u) - \alpha_i^{\text{leave}}(k_u)). \end{aligned} \quad (29)$$

The entering flow rate $\alpha_i^{\text{enter}}(k_u)$ of agent i at time step k_u is determined by the leaving flow rate of its neighbors and the external traffic demand

$$\alpha_i^{\text{enter}}(k_u) = \phi_i^1(\alpha_j^{\text{leave}}(k_u), q_p(k_u), d_i(k_u))$$

where ϕ_i^1 is a suitably defined function.

The arriving flow rate $\alpha_i^{\text{arrive}}(k_u)$ of agent i at time step k_u can be computed by its entering flow rate $\alpha_i^{\text{enter}}(k_u), \dots, \alpha_i^{\text{enter}}(k_u - \tau_i(k_u))$ during the previous time steps of k_u (including k_u)

$$\alpha_i^{\text{arrive}}(k_u) = \phi_i^2(\alpha_i^{\text{enter}}(k_u), \dots, \alpha_i^{\text{enter}}(k_u - \tau_i(k_u)))$$

where ϕ_i^2 is a suitably defined function, and $\tau_i(k_u) \in \mathbb{N}$ is an integer. The variables $\alpha_i^{\text{enter}}(k_u - t)$ can be viewed as constants at time step k_u when $t \in \mathbb{N}^+$.

The leaving flow rate $\alpha_i^{\text{leave}}(k_u)$ of agent i at time step k_u can be computed by the following equation according to (3):

$$\alpha_i^{\text{leave}}(k_u) = \phi_i^3(g_i(k_u), l_i(k_u), \alpha_i^{\text{arrive}}(k_u), n_j(k_u))$$

where ϕ_i^3 is a suitably defined function.

We define the traffic state and the control signal of urban agent i as

$$x_i^u = [n_i^T \quad l_i^T]^T, \quad u_i^u = g_i^T$$

where the superscript u of x_i and u_i denotes that it is related to the urban agent. Then, we can get the state equation of urban agent i

$$x_i^u(k_u + 1) = f_i^u(x_i^u(k_u), u_i^u(k_u), y_{ji}^u(k_u), y_{pi}^f(k_u), d_i(k_u)) \quad (30)$$

where f_i^u is a nonlinear function, $y_{ji}^u(k_u)$ contains the information on the leaving flow rate $\alpha_j^{\text{leave}}(k_u)$ and the number of vehicles $n_j(k_u)$ of urban agent j , and $y_{pi}^f(k_u)$ involves the information about the outflow $q_p(k_u)$ of freeway agent p .

2) *Freeway Agent*: Based on the partitioning method in Section IV-A, a freeway agent contains freeway links, the adjacent on-ramps, and the origin link of the freeway network.

We assume freeway link m is involved in freeway agent p , i.e., $m \in \mathcal{L}_p$.

The number of vehicles of on-ramp (i, m) that connects intersection i and link m at time step $k_f + 1$ is given as

$$n_{i,m}(k_f + 1) = \psi_{i,m}(n_{i,m}(k_f), \alpha_i^{\text{leave}}(k_f), \rho_{m,1}(k_f)) \quad (31)$$

where $\psi_{i,m}$ is a suitably defined function.

The traffic density and traffic speed of the downstream freeway segment $(m, 1)$ of on-ramp (i, m) at time step $k_f + 1$ can be given as

$$\rho_{m,1}(k_f + 1) = \psi_{m,1}^1(\rho_{m,1}(k_f), v_{m,1}(k_f), \rho_{m-1, N_{m-1}}(k_f),$$

$$v_{m-1,N_{m-1}}(k_f), n_{i,m}(k_f), \alpha_i^{\text{leave}}(k_f)) \\ v_{m,1}(k_f + 1) = \psi_{m,1}^2(\rho_{m,1}(k_f), v_{m,1}(k_f), v_m^{\text{lim}}(k_f), \\ \rho_{m-1,N_{m-1}}(k_f), v_{m-1,N_{m-1}}(k_f), n_{i,m}(k_f), \\ \alpha_i^{\text{leave}}(k_f), \rho_{m,2}(k_f))$$

where $\psi_{m,1}^1$ and $\psi_{m,1}^2$ are suitably defined functions.

If origin link o of the freeway network is adjacent to freeway link m , then the number of vehicles of link o at time step $k_f + 1$ can be given as

$$n_o(k_f + 1) = \psi_o(n_o(k_f), d_o(k_f), \rho_{m,1}(k_f)) \quad (32)$$

where ψ_o is a suitably defined function.

The traffic density and traffic speed of the downstream freeway segment $(m, 1)$ of origin link o at time step $k_f + 1$ can be described as

$$\rho_{m,1}(k_f + 1) = \psi_{m,1}^1(\rho_{m,1}(k_f), v_{m,1}(k_f), d_o(k_f), n_o(k_f)) \\ v_{m,1}(k_f + 1) = \psi_{m,1}^2(\rho_{m,1}(k_f), v_{m,1}(k_f), v_m^{\text{lim}}(k_f), d_o(k_f), \\ n_o(k_f), \rho_{m,2}(k_f)).$$

The traffic density and traffic speed of the upstream freeway segment (m, N_m) of off-ramp (m, i) that connects link m and intersection i at time step $k_f + 1$ can be given as

$$\rho_{m,N_m}(k_f + 1) = \psi_{m,N_m}^1(\rho_{m,N_m}(k_f), v_{m,N_m}(k_f), \\ \rho_{m,N_m-1}(k_f), v_{m,N_m-1}(k_f), n_{m,i}(k_f)) \\ v_{m,N_m}(k_f + 1) = \psi_{m,N_m}^2(\rho_{m,N_m}(k_f), v_{m,N_m}(k_f), v_m^{\text{lim}}(k_f), \\ v_{m,N_m-1}(k_f), \rho_{m+1,1}(k_f), n_{m,i}(k_f))$$

where ψ_{m,N_m}^1 and ψ_{m,N_m}^2 are suitably defined functions.

If freeway segment (m, t) is not adjacent to the on-ramp, the off-ramp, and the origin link, then the traffic density and traffic speed of (m, t) at time step $k_f + 1$ can be given as

$$\rho_{m,t}(k_f + 1) = \psi_{m,t}^1(\rho_{m,t}(k_f), v_{m,t}(k_f), \rho_{m,t-1}(k_f), \\ v_{m,t-1}(k_f)) \\ v_{m,t}(k_f + 1) = \psi_{m,t}^2(\rho_{m,t}(k_f), v_{m,t}(k_f), v_m^{\text{lim}}(k_f), \\ v_{m,t-1}(k_f), \rho_{m,t+1}(k_f))$$

where $\psi_{m,t}^1$ and $\psi_{m,t}^2$ are suitably defined functions.

Define the traffic state and the control signal of freeway agent p as

$$x_p^f(k_f) = [\rho_p^T(k_f) \ v_p^T(k_f) \ n_{i,m}^T(k_f) \ n_o(k_f)]^T \\ u_p^f(k_f) = [v_m^{\text{lim}}(k_f)]^T$$

where $\rho_p = [\rho_{m,1} \cdots \rho_{m,N_m} \cdots]_{m \in \mathcal{L}_p}^T$, $v_p = [v_{m,1} \cdots v_{m,N_m} \cdots]_{m \in \mathcal{L}_p}^T$, and $v_p^{\text{lim}} = [v_m^{\text{lim}} \cdots]_{m \in \mathcal{L}_p}^T$. Moreover, $(i, m) \in \mathcal{E}_{\text{on}}$ denotes the set of on-ramps in agent p . The superscript f of x_p and u_p denotes that the variables are related to the freeway agent. If there is no on-ramp or origin link in freeway agent p , the corresponding variables will be removed from the state vector.

Thus, the state equation of freeway segment p can be given as

$$x_p^f(k_f + 1) = f_p^f(x_p^f(k_f), u_p^f(k_f), y_{qp}^f(k_f), y_{ip}^u(k_f), d_o(k_f)) \quad (33)$$

where f_p^f is a suitably defined nonlinear function, $y_{qp}^f(k_f)$ contains the information on the traffic density $\rho_q(k_f)$ and the

traffic speed $v_q(k_f)$ of freeway agent q , and $y_{ip}^u(k_f)$ involves the information about the leaving flow rate $\alpha_i^{\text{leave}}(k_f)$ and the number of vehicles $n_i(k_f)$ of urban agent i .

C. Multiagent Control Problem Formulation

According to the partitioning method in Section IV-A, the objective function and the constraints of the optimization problem of the mixed traffic network can be decomposed into each agent. Thus, the suboptimization problem in each agent can be formulated as follows.

1) *Urban Agent*: The optimization problem of urban agent $i \in \mathcal{A}_u$ can be formulated as

$$\min_{\tilde{u}_i^u(k)} J_{i,\text{local}}(k_c) = T_u \sum_{k=k_u^0}^{k_u^1} e_i^T n_i(k + 1) \\ + \sum_{k=k_c}^{k_c+N_p-2} (u_i^u(k + 1) - u_i^u(k))^T R_i (u_i^u(k + 1) - u_i^u(k)) \\ \text{s.t. Eq. (25), (27), (30)}$$

where $e_i \in \mathbb{R}^{|\mathcal{E}_i|}$ is a unit vector of dimension $|\mathcal{E}_i|$, $|\mathcal{E}_i|$ is the number of roads in agent i , and C_i is the vector consisting of the storage capacity of roads in agent i .

2) *Freeway Agent*: The optimization problem of freeway agent $p \in \mathcal{A}_f$ can be formulated as

$$\min_{\tilde{u}_p^f(k)} J_{p,\text{local}}(k_c) \\ = T_f \sum_{k=k_f^0}^{k_f^1} \left(n_o(k + 1) + \sum_{(i,m) \in \mathcal{E}_{\text{on}}, m \in \mathcal{L}_p} n_{i,m}(k + 1) \right. \\ \left. + \sum_{m \in \mathcal{L}_p} \sum_{t=1}^{N_m} L_m \lambda_m \rho_{m,t}(k + 1) \right) \\ + \sum_{k=k_c}^{k_c+N_p-2} (u_p^f(k + 1) - u_p^f(k))^T R_p (u_p^f(k + 1) - u_p^f(k)) \\ \text{s.t. Eq. (26), (28), (33).}$$

Therefore, in the distributed control architecture, each agent solves its own optimization problem by minimizing the local objective function subject to the local constraints and local dynamics. The agent senses its local variables at each time step, gets the interaction variables from its neighbors through communication, and then computes the control signals of the agent based on the obtained information. Note that the interaction variables between the agent and its neighbors are also involved in the optimization problem, which needs to be suitably estimated. In Section IV-D, we will discuss how to deal with the interaction variables.

D. Multiagent MPC

In the distributed control of the mixed traffic network, the interaction variables between the agents are also involved in the suboptimization problem of each agent. In this article, the agents try to seek the Nash equilibrium of the control system. Since the iteration computation of the agents are not

necessarily converged, the number of the iteration computation is restricted by $S_{\max}$.

The optimization-communication steps for computing the control signals of agent $i \in \mathcal{A}_u \cup \mathcal{A}_f$ at time step k_c can be described as follows.

- 1) Obtain the local state $x_i(k_c)$ of agent i at the current time step, set the iteration counter $s = 0$, and assume that the interaction variables over the prediction horizon $\tilde{y}_{ij}^{(s)}(k_c)$ between i and neighboring agent j at iteration s are equal to the actual value $\tilde{y}_{ij}(k_c - 1)$ at last time step.
- 2) Communicate with neighboring agent j of i , and transmit the interaction variables $\tilde{y}_{ij}^{(s)}(k)$ and get $\tilde{y}_{ji}^{(s)}(k)$.
- 3) Set the estimated coupling variables $\tilde{z}_{ji}^{(s)}(k_c)$ between agent j and i : $\tilde{z}_{ji}^{(s)}(k_c) = \tilde{y}_{ji}^{(s)}(k_c)$.
- 4) Solve the optimization problem of agent i , and obtain the optimal control sequence $\tilde{u}_i^{(s+1)}(k_c)$ and the predicted interaction variables $\tilde{y}_{ij}^{(s+1)}(k_c)$ over the prediction horizon

$$\begin{aligned} & (\tilde{u}_i^{(s+1)}(k_c), \tilde{y}_{ij}^{(s+1)}(k_c)) \\ &= \underset{\tilde{u}_i(k_c), \tilde{y}_{ij}(k_c)}{\operatorname{argmin}} \left(J_{i,\text{local}}(x_i(k_c), \tilde{u}_i(k_c), \tilde{y}_{ij}^{(s+1)}(k_c), \tilde{z}_{ji}^{(s)}(k_c)) \right. \\ & \quad \left. + J_{i,\text{inter}}(x_i(k_c), \tilde{u}_i(k_c), \tilde{y}_{ij}^{(s)}(k_c), \tilde{y}_{ij}^{(s+1)}(k_c)) \right) \end{aligned}$$

where $J_{i,\text{inter}}(k)$ is added to the local objective function of i to accelerate the convergence speed of the iteration

$$J_{i,\text{inter}} = (\tilde{y}_{ij}^{(s+1)}(k) - \tilde{y}_{ij}^{(s)}(k))^T P_i (\tilde{y}_{ij}^{(s+1)}(k) - \tilde{y}_{ij}^{(s)}(k)) \quad (34)$$

with P_i being the suitably defined matrix. Note, however, that other methods that are helpful for the convergence can also be used.

- 5) Check whether the termination condition of the iteration for seeking the Nash equilibrium is satisfied

$$|\tilde{y}_{ij}^{(s+1)}(k_c) - \tilde{y}_{ij}^{(s)}(k_c)| < \epsilon \text{ or } s > S_{\max} \quad (35)$$

where ϵ is a small positive tolerance, and $S_{\max}$ is the maximum number of iterations. If it is not satisfied, increase the iteration counter s , i.e., let $s = s + 1$, and go to step 2); otherwise, go to step 6).

- 6) Set the optimal control sequence $\tilde{u}_i^*(k_c) = \tilde{u}_i^{(s+1)}(k_c)$, and apply the first sample of $\tilde{u}_i^*(k_c)$ to the system. Then, go to step 1) to move on to the next time step.

In the proposed distributed control approach, the subsystem tries to achieve the Nash equilibrium by solving the suboptimization problem iteratively. In the cases where the convergence is obtained by the agents, the Nash equilibrium can be achieved. In the cases where the coordination between agents is terminated by the limitation of the upper bound of iterations, a suboptimal solution is achieved as the subsystem always attempts to minimize its objective function.

V. CASE STUDY

To validate the effectiveness of the distributed control strategy, it is applied to a mixed traffic network through a

numerical experiment. The mixed network is extracted from an area in Shanghai, China, as shown in Fig. 4. The proposed control scheme can be easily extended to other road networks with the distributed architecture. The simulation is carried out by the open-source microscopic road traffic simulation package ‘‘Simulation of Urban MObility’’ (SUMO).

A. Road Network and Setup

The mixed road network in the case study consists of an urban network and a freeway network that are connected by two on-ramps and two off-ramps, as shown in Fig. 4. The urban network includes 15 intersections, 13 origin and destination nodes, and 33 two-way roads ranging from 230 to 750 m. The concrete parameters about the length and width of each road can be found in the Google Map by searching their names. The cycle time of the traffic lights in each intersection is $T_u = 60$ s, which is also the sampling time in the urban network. The saturated flow rate is $\mu = 0.8$ veh/s, the average vehicle length is $l_{\text{veh}} = 5$ m, and the free flow speed is $v_{u,d}^{\text{free}} = 13$ m/s for all roads. In addition, the lower and upper bounds for the green time duration are $G_{\min} = 10$ s and $G_{\max} = 50$ s, respectively. The prediction time horizon is $N_p = 10$.

The freeway network consists of six freeway links of 800 m each and an origin link that lies upstream of link 1. Each freeway link is equipped with a variable message sign where the speed limit is displayed. The number of segments and the number of lanes are $N_m = 2$ and $\lambda_m = 3$, respectively, and the free flow speed is $v_m^{\text{free}} = 104$ km/h for all freeway links. The jam density $\rho_m^{\max} = 170$ veh/lane/km and the critical density $\rho_m^{\text{crit}} = 36$ veh/lane/km are identified from the simulation. Moreover, the sampling time interval $T_f = 10$ s, the time constant $\tau = 18$ s, and $\kappa = 40$ are from [32]. In the implementation of the VSL scheme, the lower bound of VSL is $v_{\min}^{\text{lim}} = 30$ km/h, and the maximum decrease is $\Delta v_{\max} = 20$ km/h.

In the proposed distributed scheme, the control time interval is $T_c = 60$ s. The urban network is partitioned into 15 urban agents, each consisting of an intersection. The freeway network is partitioned into two freeway agents. Freeway agent 1 is composed of the freeway origin link, freeway segments 1–6, on-ramp 1, and freeway agent 2 is composed of freeway segments 7–12 and on-ramp 2.

B. Compared Schemes

To demonstrate the improvement of the control performance introduced by the cooperative control of the freeway and the urban network, the proposed cooperative distributed (CD) control scheme is compared with the non-CD (NCD) control scheme where the urban and freeway networks are controlled independently. Moreover, to evaluate the computational efficiency and the degradation of the control performance of the distributed control scheme, the CD control scheme is further compared with the CC scheme where the optimization problem of the mixed network is solved by a centralized unit.

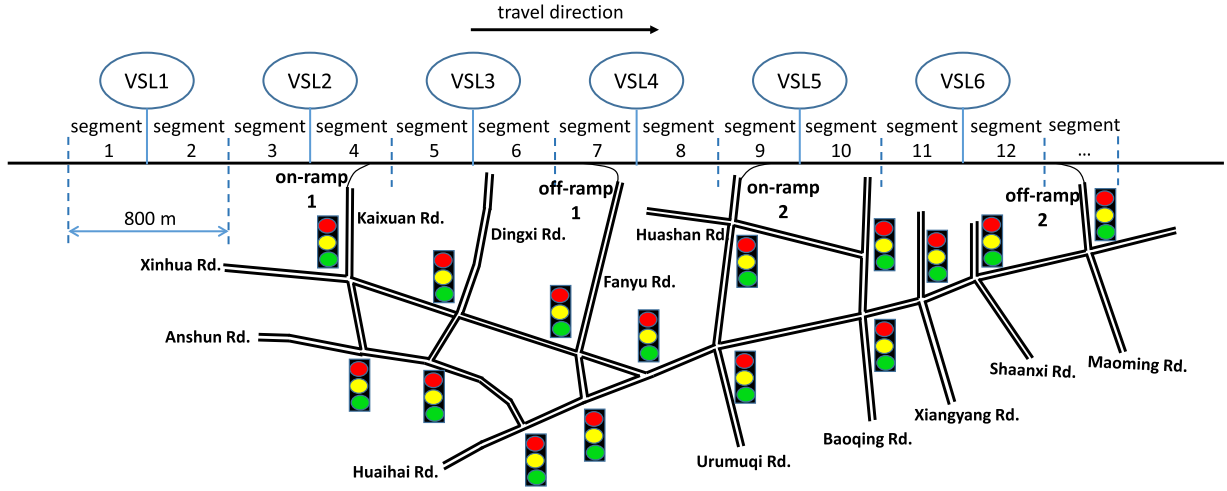


Fig. 4. Mixed traffic network in Shanghai, China.

For the comparison of the performance of different control strategies, the following criteria are collected by SUMO (provided by SUMO directly) for assessments.

- 1) *TTS*: The TTS of vehicles in the whole network.
- 2) *Waiting Time (WT)*: The total time in which vehicles are considered stopped (with a speed of less than 0.1 m/s). It characterizes the time spent by the standing vehicles in the whole network.
- 3) *CPU Time*: The average optimization CPU time.

C. CPU Time

With the decomposition of the mixed traffic network, the computational complexity of the distributed control is greatly reduced compared with the CC scheme.

The computational complexity of the optimization problem of the mixed traffic network is proportional to the scale of the network, including the number of intersections in the urban network and the number of freeway links in the freeway network. In the CC strategy, the overall optimization problem is solved by a centralized controller. In contrast, by decomposing the traffic network into N subnetworks with only one intersection in each urban subsystem and a small number of freeway links in each freeway subsystem, the overall optimization problem can be decomposed into N optimization problems with fewer control variables in each subsystem, which can be computed in parallel. As the computational complexity of the suboptimization problem is proportional to the scale of the subnetwork, the total amount of calculation in the distributed scheme can be much less than that in the centralized strategy even taking account into the coordination process between the subnetworks.

In the simulation, the mixed traffic network is decomposed into 15 urban agents and six freeway agents in the CD and the NCD schemes. The optimization problem of the urban agent is an MILP problem and is solved by the CPLEX solver embedded in MATLAB. The optimization problem of the freeway agent is a nonlinear problem and is solved by the sequential quadratic programming (SQP) algorithm in

TABLE IV
CPU TIME OF DIFFERENT CONTROL SCHEMES

strategy	average CPU time (s)	maximum CPU time (s)
CD	5.07	37.33
NCD	4.83	32.62
CC	141.92	452.73

MATLAB. Since the solution of the nonlinear problem is related to the location of the initial point, five initial points are used, and the optimal solution with the minimum objective value is implemented. The Nash equilibrium is pursued by the agents in the distributed control scheme to achieve a coordinated solution. The maximum number of iterations is set to 5.

In the CC scheme, the overall optimization problem is a nonlinear problem, in which the control variables consist of traffic signals of 15 intersections and variable speed limits of six freeway links and is solved by the SQP algorithm in MATLAB. Similarly, five initial points are used. All the computations are conducted on a computer with a 3.6-GHz Intel Core Processor and 32-GB RAM.

The average and the maximum CPU time with different control strategies are shown in Table IV. The maximum CPU time of the CC scheme CC has exceeded 400 s, which is reduced by two orders of magnitude by the distributed control schemes CD and NCD. Therefore, the computational efficiency of the CC scheme can be significantly improved by the distributed control strategy. As the maximum CPU time of the CD scheme is less than 40 s, it can be implemented in real-time control of the mixed traffic network.

D. Simulation Results

The control performance of the control schemes is assessed under scenario 1 with the light external traffic demand and scenarios 2 and 3 with the heavy external traffic demand. Moreover, the transfer flow among the urban and freeway networks is heavier in scenario 3 compared with scenario 2,

TABLE V
SUMMARY OF THE SIMULATION RESULTS

Scenario	strategy	TTS ($\times 10^3$ veh-h) improvement	WT ($\times 10^3$ veh-h) improvement
1	CD	1.78 (-7.3 %)	0.90 (-11.7 %)
	CC	1.59 (-17.2 %)	0.82 (-19.6 %)
	NCD	1.92 (-)	1.02 (-)
2	CD	2.36 (-26.5 %)	1.04 (-33.8 %)
	CC	1.92 (-40.2 %)	0.94 (-40.1 %)
	NCD	3.21 (-)	1.57 (-)
3	CD	2.78 (-55.7 %)	1.36 (-62.9 %)
	CC	2.12 (-66.2 %)	1.12 (-69.5 %)
	NCD	6.28 (-)	3.67 (-)

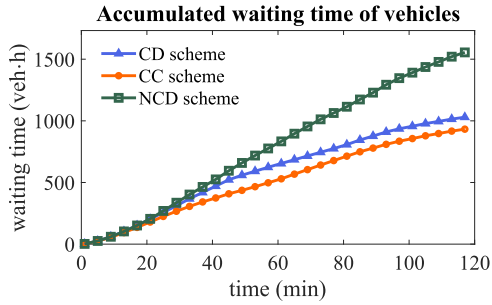


Fig. 5. Accumulated WT of vehicles in the network under Scenario 2.

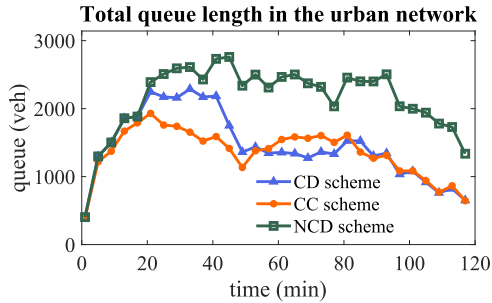


Fig. 6. Total queue length in the urban network under Scenario 2.

resulting in stronger couplings between the two parts. The simulation results are shown in Table V.

In scenario 1, both the TTS and the WT under the NCD scheme are slightly reduced by the proposed CD scheme in the light traffic condition.

In scenario 2, the transfer flow between the urban and freeway parts is enlarged due to the increasing external traffic demand. The superiority of the CD scheme becomes clearer compared with scenario 1. The TTS and the WT under the NCD scheme are reduced by 26.5% and 33.8% by the CD scheme.

The evolution of the accumulated WT of vehicles of the whole network in scenario 2 is presented in Fig. 5. The WT under the NCD scheme is significantly reduced by the CD scheme and the CC scheme. Moreover, the evolution of the total queue length in the urban network is shown in Fig. 6. The queue length under the CD and the NCD schemes is increased since the 20th time step. However, the queue length under the CD scheme is reduced to the approximate level with the CC

scheme since the 40th time step, while the queue length under the NCD scheme remains high until the 100th time step.

To explore the effects of the urban congestion on the freeway network, the queue length of off-ramp 2 and the traffic density of freeway links are presented in Figs. 7 and 8, respectively. Under the CD scheme, the queue length of off-ramp 2 and the traffic density of its upstream freeway link (freeway segment 12) is increased after the 20th time step. Then, the congestion on these roads is relieved after the 40th time step, which is consistent with the change of the queue length in the urban network in Fig. 6.

The evolution of the VSL values for freeway segments 7–8, 9–10, and 11–12 (freeway links 4, 5, and 6) and the green time duration of the phase that corresponds to the outflow of off-ramp 2 (phase 2) are given in Figs. 9 and 10, respectively. From the 20th to 40th time steps during which the queue length of off-ramp 2 is increased, the VSL control of freeway segments 7–12 in the CD scheme is activated. The speed limits of the freeway links are decreased with the maximum allowable speed $\Delta v = 20$ km/h until it reaches the minimum value $v_{\min}^{\lim} = 30$ km/h. Besides, the average green time of phase 2 in which the vehicles of off-ramp 2 have right of way is also longer compared with other periods, revealing the cooperative control of the VSL strategy in the freeway network and the traffic signal control in the urban network in the CD scheme. The restricted inflow from the freeway network caused by speed limits contributes to the relief of urban congestion after the 40th time step. Benefit from the urban congestion alleviation, the congestion on off-ramp 2 and its upstream freeway links is also mitigated.

In contrast, the speed limits of the freeway links and the green time duration of phase 2 in the NCD scheme present large fluctuations due to the noncooperative control. As a consequence, the queue length of off-ramp 2 in Fig. 7(b) and the traffic density of its upstream freeway links in Fig. 8(b) also present oscillations since the 20th time step, leading to the recurrent congestion in the freeway network implied by the high traffic density. The difference between the simulation results of the CD and the NCD control schemes reveals that the cooperative control of the urban and freeway networks can restrain the congestion propagation in the mixed traffic network.

When the transfer flow between the urban and freeway networks is increased in scenario 3, the benefits of the cooperative control of the two parts are more obvious compared with scenario 2. In addition, the performance degradation of the CD scheme compared with the CC scheme is larger compared with scenario 2. The TTS and the WT under the NCD scheme are reduced by 55.7% and 62.9% by the CD scheme.

The evolution of queue length of off-ramp 2 and the traffic density of freeway links are shown in Figs. 11 and 12, respectively. When the queue length of off-ramp 2 is increased after the 30th time step in the NCD scheme, the congestion spreads to its upstream freeway links and leads to severe congestion propagation throughout the freeway network. Due to the stronger couplings between the urban and freeway networks in scenario 3, the spill-back effects of the urban

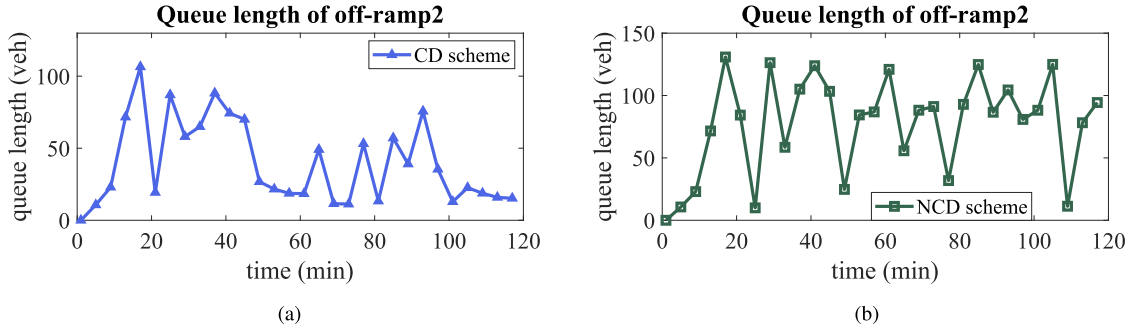


Fig. 7. Queue length of off-ramp 2 with different control strategies under Scenario 2. (a) CD scheme. (b) NCD scheme.

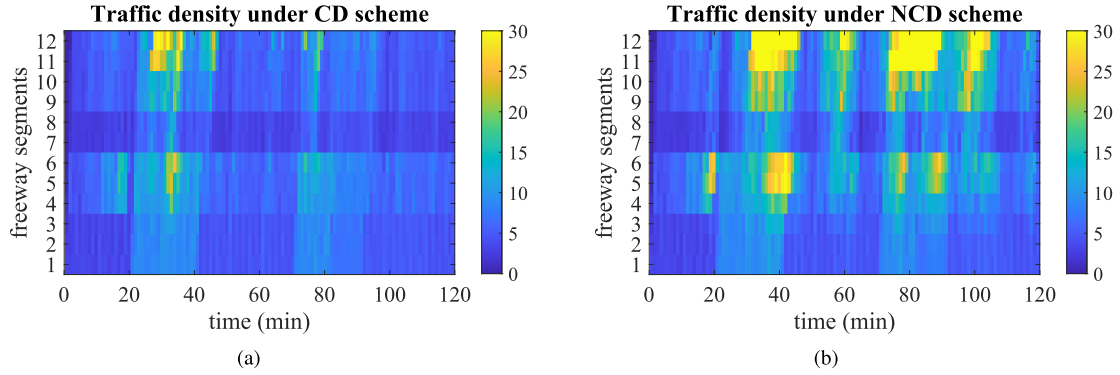


Fig. 8. Traffic density of the freeway network with different control strategies under Scenario 2. (a) CD scheme. (b) NCD scheme.

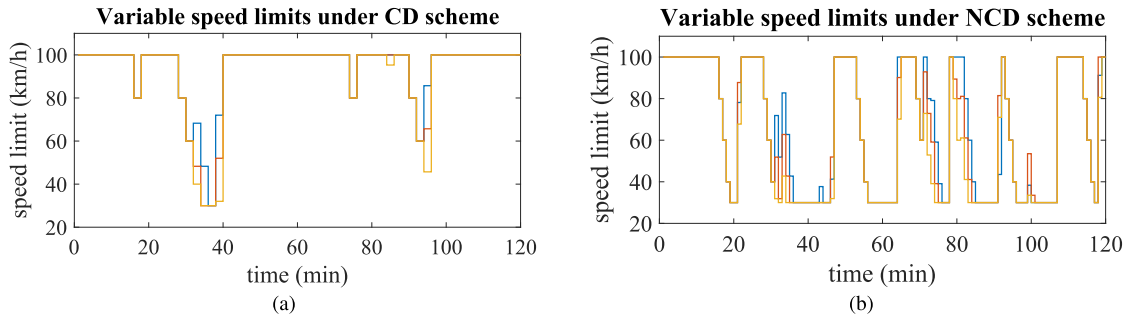


Fig. 9. Variable speed limits of freeway links with different control strategies under Scenario 2. (a) CD scheme. (b) NCD scheme.

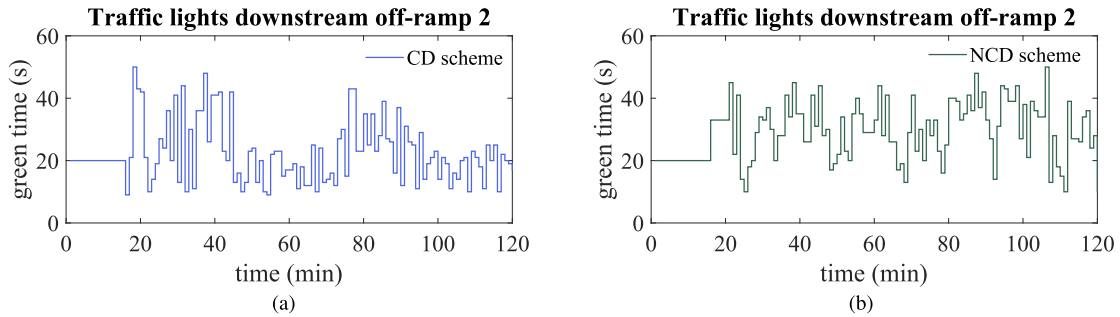


Fig. 10. Green time of phase 2 with different control strategies under Scenario 2. (a) CD scheme. (b) NCD scheme.

congestion are more profound and protracted. In comparison, though the queue length of off-ramp 2 increases after the 10th and 40th time steps in the CD scheme, the congestion

propagation to the freeway network is efficiently inhibited, devoting to the reduction of the TTS and WT in the whole network.

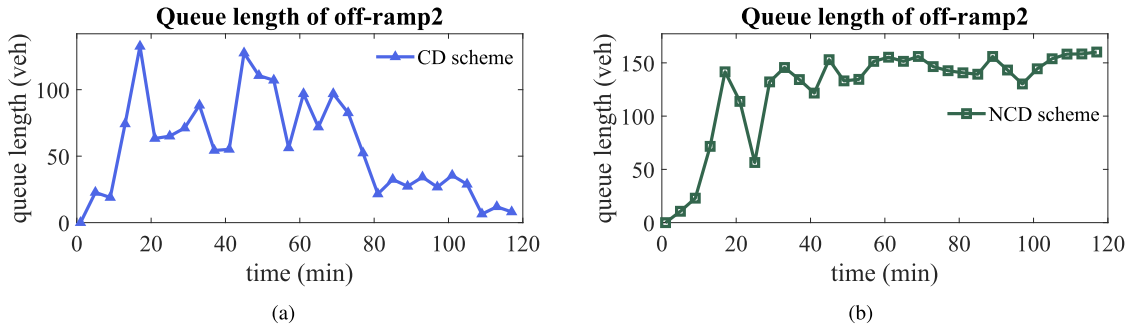


Fig. 11. Queue length of off-ramp 2 with different control strategies under Scenario 3. (a) CD scheme. (b) NCD scheme.

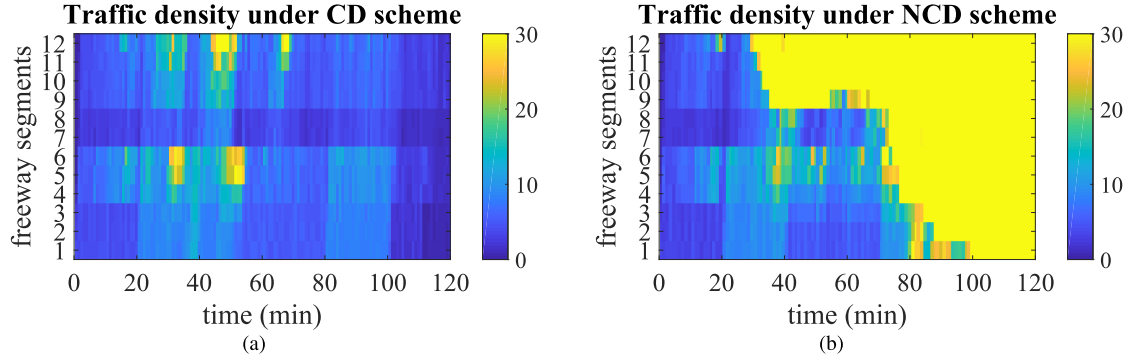


Fig. 12. Traffic density of the freeway links with different control strategies under Scenario 3. (a) CD scheme. (b) NCD scheme.

Above all, the mixed traffic network can benefit a lot from the cooperative control of the urban and freeway networks, especially in the strongly coupled cases.

VI. CONCLUSION

In this article, a distributed control framework is proposed for the cooperative control of a mixed transportation network consisting of the freeway and urban networks. First, the traffic flow model of the urban and freeway networks is presented. To incorporate the spill-back effects of off-ramps, the model is extended by considering the influence of the traffic condition of off-ramps on the number of lanes and the traffic outflow of the upstream freeway link. In the cooperated control of the mixed traffic network, a model predictive control strategy is applied to optimize the traffic signals in the urban network and the variable speed limits in the freeway network. To reduce the computational and the communicational burden of a centralized controller, a distributed control framework is proposed. Then, the overall optimization problem can be decomposed into a number of suboptimization problems, which are solved by each agent. At each time step, the agents compute the control signals to seek the Nash equilibrium of the system. Finally, the control performance of the proposed distributed control strategy of mixed traffic networks is evaluated by a numerical example.

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